

Territorial Analysis and Lithium Exploration Potential in Quebec Based on SIGÉOM Data

An exploratory study using Python, QGIS, and SIGÉOM open data

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Tools used: Python — QGIS — SIGÉOM

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1. Introduction

The energy transition has placed lithium at the center of North American and European industrial strategies. Used in the manufacture of electric vehicle batteries and stationary electricity storage, this metal is the subject of growing demand that is pushing several mining jurisdictions to reassess their potential. Quebec is among the North American territories where lithium exploration has accelerated in recent years, particularly in the James Bay region and certain sectors of Abitibi, two areas associated with granitic pegmatite-type geological formations that are generally favorable to this type of mineralization.

Quebec has a particular advantage for studying this dynamic: the SIGÉOM database, managed by the Ministère des Ressources naturelles et des Forêts, provides the public with a very detailed set of information on mining projects, geological indices, and exploration targets recorded across the territory. This data openness makes it possible to build, using relatively simple tools, an initial spatial reading of lithium-related activity in Quebec, without resorting to field-based geological expertise.

This work follows that logic. It does not claim to produce a professional geological assessment of mining potential, but rather proposes an exploratory analysis combining data processing, mapping, and an economic-geography reading. The objective is to show how the SIGÉOM databases can be cross-referenced with one another in order to locate existing lithium projects, identify lithium-related mineralized occurrences beyond the projects already known, rank these occurrences using an exploratory score, and then compare them with the exploration targets recorded in the same database.

The research question structuring this analysis is the following: to what extent do the lithium projects recorded in Quebec reveal a territorial concentration, and how does cross-referencing with GMLI occurrences and exploration targets help identify areas of interest for exploratory prospecting?

To answer this question, the report first presents the data and methodology used, then successively details the identification of lithium projects, the analysis of GMLI occurrences, the construction of the exploratory score, the cross-referencing with exploration targets, and the resulting final mapping. The limitations of the approach are then discussed before the conclusion.

2. Data, Tools, and General Method

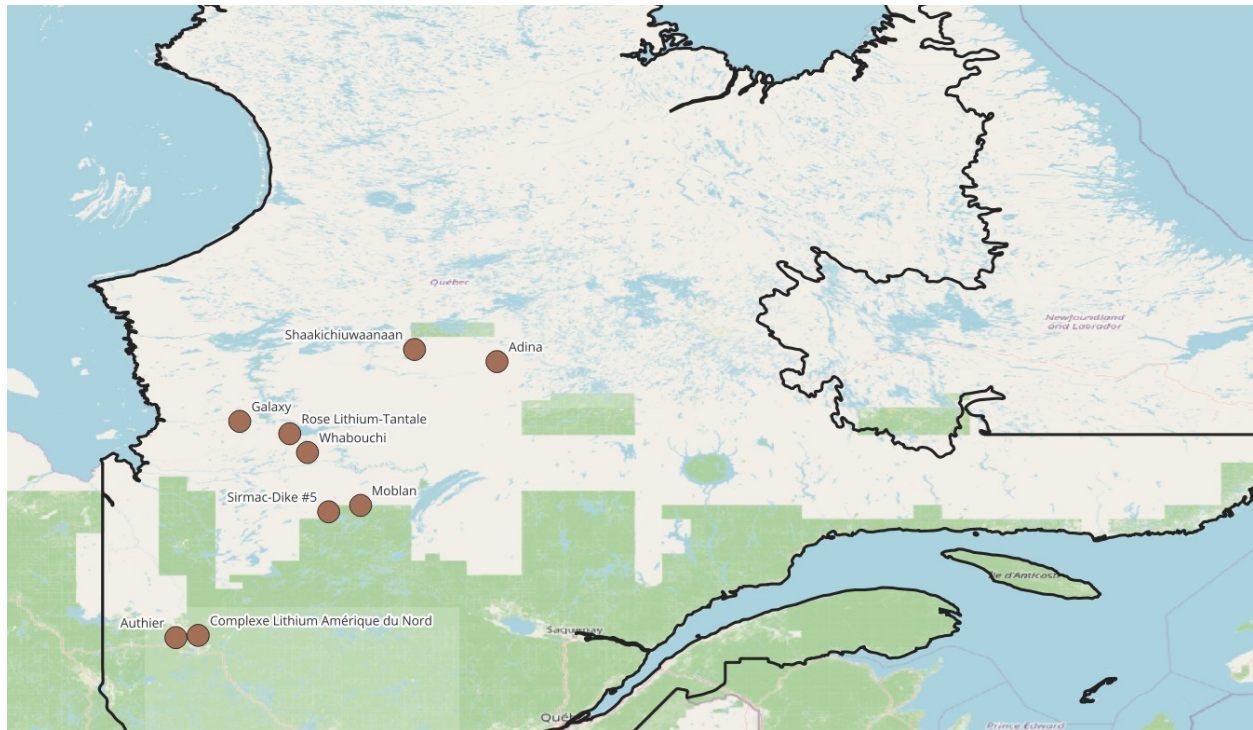
The analysis relies on three public databases from SIGÉOM: the “Mines and Projects” database, which records mining projects at various stages of progress; the “Metallic Substances” database, which gathers mineralized occurrences identified across the territory, regardless of their project status; and the “Exploration Targets” database, which brings together sectors that have been flagged by geologists, without this flagging necessarily constituting confirmation of an exploitable resource.

Data processing was carried out using Python, which was used to filter substance columns, extract relevant occurrences, build the exploratory score, calculate distances between points, and export the summary files used in this report. The mapping was produced in QGIS, based on an OpenStreetMap basemap and, for the general location map, a Natural Earth layer used to overlay Quebec's administrative boundary.

It is important to clarify from the outset the status of this approach. None of the steps described below constitutes geological proof of the existence of an exploitable deposit. The work aims to organize and cross-reference already public data in order to highlight areas of interest, that is, sectors where several favorable signals overlap spatially. These results should be read as a screening aid tool, capable of guiding further investigation, and not as a conclusion on the actual mining value of the sites identified.

3. Identification of Recorded Lithium Projects

The first step of the analysis consisted of querying the “Mines and Projects” database using the substance columns, searching for the Li₂O code, which designates lithium oxide used as a marker of lithium content in geological reports. This filtering identified nine projects spread across Quebec's territory: Complexe Lithium Amérique du Nord, Galaxy, Rose Lithium-Tantale, Moblan, Whabouchi, Authier, Sirmac-Dike #5, Shaakichiuwaanaan, and Adina.



Map 1 — General location of the 9 lithium projects recorded in SIGÉOM's “Mines and Projects” database.

The spatial distribution of these nine projects is not uniform. Seven of them are located within the boundaries of the Plan Nord — namely Galaxy, Rose Lithium-Tantale, Moblan, Whabouchi, Sirmac-Dike #5, Shaakichiuwaanaan, and Adina — while only two projects lie outside this perimeter: Complexe Lithium Amérique du Nord and Authier, both located in Abitibi.

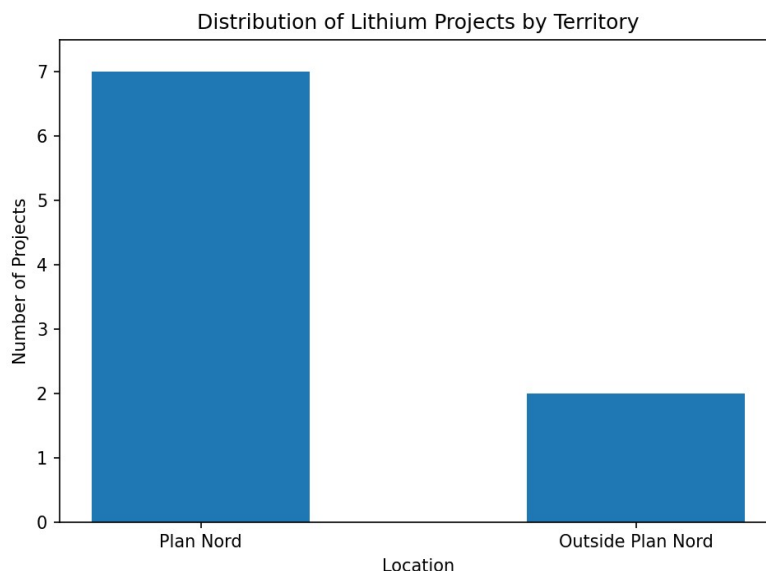


Chart 1 — Distribution of lithium projects according to their location within or outside the Plan Nord.

This concentration within the Plan Nord is not a statistical coincidence. The Plan Nord constitutes a territorial framework that has structured a significant share of Quebec's mining activity for more than a decade, by organizing access to infrastructure, energy development, and, more broadly, the conditions of accessibility to territories that are otherwise remote from major transportation routes. The predominant presence of lithium projects within this perimeter illustrates a classic logic in economic geography: the location of mining activities does not depend solely on the nature of the resource sought, but also on the institutional and logistical environment that makes this resource accessible and exploitable. Here, the territory functions as a genuine factor of economic organization, rather than as a simple neutral backdrop on which projects would be distributed at random.

Examination of the promoters associated with these nine projects confirms a certain diversity of actors, although one promoter, Elevra Lithium Ltd., holds three of the nine recorded projects. The other six projects each belong to a distinct promoter: Rio Tinto Lithium Canada Inc., Corporation Lithium Éléments Critiques, Nemaska Lithium Inc., Vision Lithium Inc., Ressources PMET Inc., and Li-FT Power Ltd.

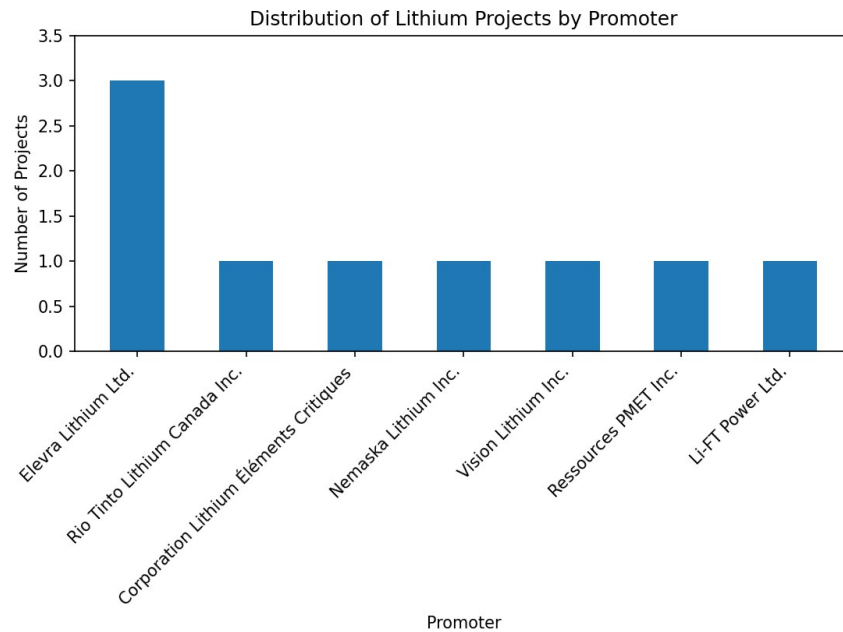


Chart 2 — Distribution of lithium projects by promoter.

This relatively concentrated structure around a single promoter resembles a form of polarization of actor strategies around certain mining areas already identified as promising. An actor holding several projects in the same region tends, in effect, to reinforce that area's territorial specialization around the lithium sector, by concentrating investment, infrastructure, and accumulated expertise there. The SIGÉOM database also contains a status field for each project, but the exact meaning of these codes could not be fully documented at this stage; any reading of the distribution by status should therefore be treated with caution and is not developed further in this report.

4. Analysis of GMLI Occurrences

The nine projects identified previously represent only a fraction of the information available on lithium in Quebec. To broaden the analysis, SIGÉOM's "Metallic Substances" database was used. This database records a much larger set of mineralized bodies, comprising 9,291 rows across 93 columns, which are not limited to projects under development but also include indices and showings at very different stages of progress.

Python-based filtering was used to extract, based on the GMLI code, the full set of occurrences associated with lithium. This extraction produced 159 occurrences, exported to the file `indices_lithium_gmli_quebec.csv`. It is important to distinguish these occurrences from the projects recorded in the previous chapter: a GMLI occurrence may correspond to a simple indication with no documented work, an indication that has been the subject of exploration work, or a showing for which a resource has been partially estimated. The summary provided indicates that, of the 159 occurrences, 85 correspond to an indication with no work, 61 to a worked indication, 12 to a showing, and only one to an active mine.

This distribution confirms that the vast majority of GMLI occurrences remain at an exploratory or weakly documented stage. Their relevance to this report lies precisely in this: they make it possible to broaden the spatial analysis beyond sites already engaged in a project, and to observe whether the areas where these occurrences are concentrated overlap with sectors already identified by industry.

5. Construction of the Exploratory Score

To rank the 159 GMLI occurrences according to their potential interest, an exploratory score was built from several cumulative criteria: the presence of the GMLI code itself, the fact that lithium is the main substance of the occurrence, membership in a pegmatite-type typology, the existence of an estimated resource, the state of the mineralized body (showing or worked indication), the presence of substances generally associated with lithium such as tantalum, cesium, beryllium, niobium, or rubidium, and finally geographic proximity to an already known lithium project.

The nature of this indicator must be emphasized: it is in no way an official measure nor geological proof. The score serves only to rank occurrences relative to one another, based on observable signals in the database, in order to direct attention toward the sectors that combine the largest number of favorable characteristics.

Potential Class	Number of Occurrences
High potential	102
Medium potential	55
Low potential	2
Total	159

Table 1 — Distribution of GMLI occurrences by potential class.

The result shows a clear majority of occurrences classified as high potential, with 102 out of 159, compared to 55 at medium potential and only 2 at low potential. This distribution is largely explained by the very nature of the sample: the occurrences were pre-selected based on the GMLI criterion, meaning they already share several of the characteristics sought by the score. The typology associated with granitic pegmatites dominates very widely, with 148 occurrences out of 159, corresponding to the most common type of lithium mineralization in Quebec's geological context.

Examination of the ranking shows that the ten occurrences obtaining the maximum score are, in almost every case, located in the immediate vicinity of an already recorded lithium project: Cyr-Lithium near Galaxy, Lac Moblan Ouest near Moblan, Rose near Rose Lithium-Tantale, Authier near Authier, Sirmac near Sirmac-Dike #5, Whabouchi near Whabouchi, and CV5, CV13, and Adina Main near Shaakichiuwaanaan and Adina respectively. This result is not surprising in itself, since proximity to a known lithium project is one of the criteria used in calculating the score. This correspondence does, however, provide an indication of internal consistency for the method: the highest-ranked occurrences are often located in sectors already structured around lithium projects. This does not scientifically validate the score, but shows that it captures signals already partially documented in the SIGÉOM data.

6. Cross-Referencing with Exploration Targets

The third database used is SIGÉOM's "Exploration Targets" database, which gathers 5,622 targets described across 105 columns. An exploration target corresponds to a sector that has been flagged by geologists as being of interest for further work, without this flagging being equivalent to confirmation of a deposit.

An initial filtering attempt using the keyword "Li" proved unsuitable: this string of letters appears in many terms unrelated to lithium, for example "applied" or "population" (in their French equivalents), generating a high number of false positives. The filtering was therefore reformulated based on a list of more specific keywords: lithium, pegmatite, spodumene, tantalum, cesium, rubidium, and niobium. This methodological correction made it possible to retain 44 exploration targets actually linked to lithium or pegmatites, exported to the file `cibles_exploration_lithium_pegmatite.csv`. These 44 targets obviously do not constitute proof of a deposit; they are merely compatible, by their descriptive content, with a lithium exploratory prospecting approach.

A spatial cross-referencing was then carried out between the 159 scored GMLI occurrences and the 44 targets thus retained. For each occurrence, the nearest target was identified and the corresponding distance was calculated in kilometers, then sorted into four proximity classes.

Proximity Class	Number of Occurrences
Very close to a target	41
Close to a target	12
Within a 100 km radius	38
Far from targets	68
Total	159

Table 2 — Proximity of GMLI occurrences to exploration targets.

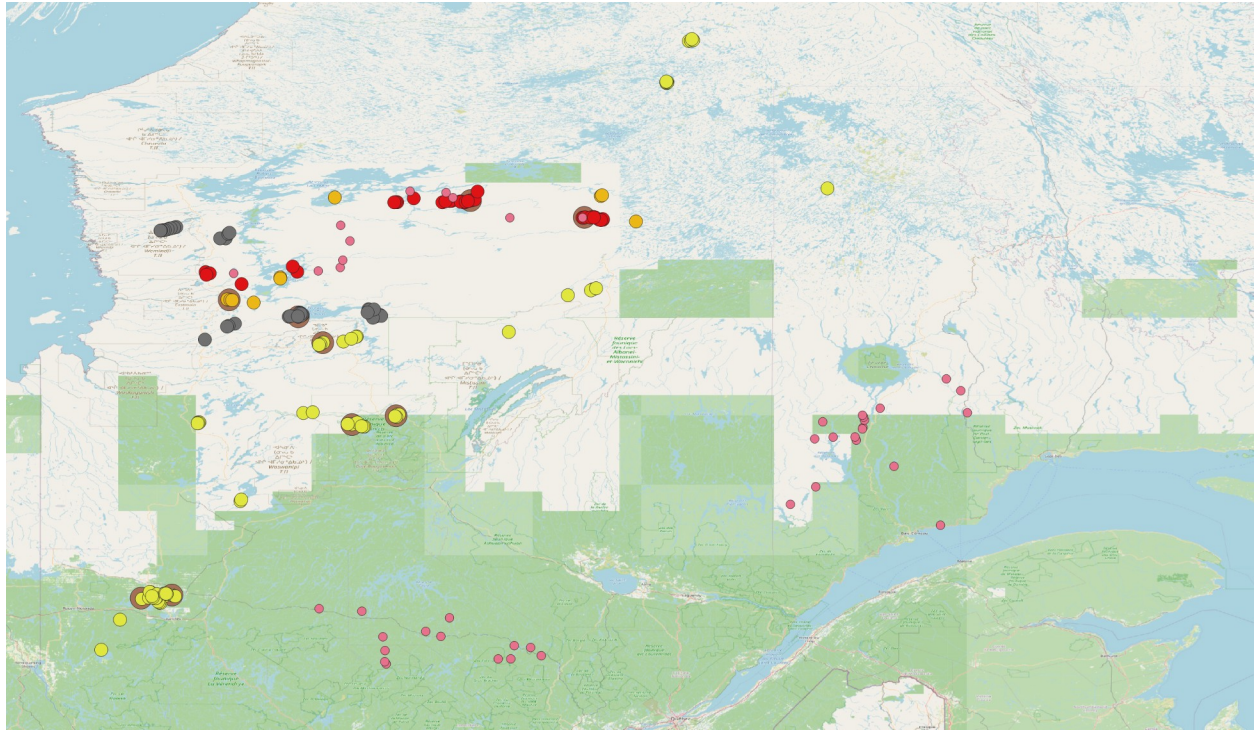
In total, 91 out of 159 occurrences — slightly more than half the sample — are located less than 100 km from an exploration target linked to lithium or pegmatites, combining the "very close," "close," and "within a 100 km radius" classes. This proportion suggests that the occurrences identified by the score are not randomly scattered across the territory, but largely fall within sectors already flagged by previous exploration work.

The combined ranking of score and distance to targets highlights a group of occurrences of particular interest for an exploratory reading. Adina Main, already classified as high potential, is located about 1 km from the "Adina" target; CV13 and CV5 are located respectively about 12, 15, 16, and 17 km from target 10-SB-6326; Cyr-Lithium is located a little over 38 km from the "Béryl" target. Other occurrences classified as high potential by the score, however, such as Whabouchi, Authier, or Lac Moblan Ouest, are found to be relatively far from any recorded

target, at distances exceeding 100, or even 200 km. This contrast illustrates an important limitation of the approach: a high score based on the intrinsic characteristics of an occurrence is not necessarily accompanied by proximity to a documented exploration target, which may be due as much to the actual history of local exploration as to gaps in the coverage of the targets database. The full cross-referenced ranking is provided in the appendix.

7. Final Cartographic Analysis

The final map overlays all the elements used in this report: the nine known lithium projects, the 159 GMLI occurrences classified by their proximity to an exploration target, and the 44 retained exploration targets, on an OpenStreetMap basemap.



Map 2 — Cross-referencing of lithium projects, GMLI occurrences, and exploration targets associated with lithium or pegmatites.

The symbology used distinguishes occurrences according to their proximity class to a target: red points indicate occurrences very close to a target, orange points those that are close, yellow points those located within a 100 km radius, and gray points the distant occurrences. Known lithium projects appear as large brown dots, while exploration targets are represented by small pink dots.

Reading this map reveals two main areas of overlap. The first is located in the northwestern sector of the study area, around the Galaxy, Rose Lithium-Tantale, and Whabouchi projects, where occurrences classified as very close to a target are concentrated alongside several exploration targets that are themselves clustered together. The second area is located further south, in the Authier and Complexe Lithium Amérique du Nord sector, where a concentration of yellow and gray occurrences coexists with already established projects, without a nearby exploration target necessarily reinforcing this signal. Between these two poles, an alignment of small pink points in the eastern part of the map corresponds to exploration targets for which few GMLI occurrences have, so far, been recorded in their immediate vicinity.

This spatial configuration can be read through the lens of territorial polarization: prospecting activity is not evenly distributed across Quebec's territory, but is organized around a few poles where the combination of several signals — high score, proximity to a project, and proximity to a target — reinforces the sector's interest. The northwestern sector, already partly included within the Plan Nord perimeter, illustrates how territorial accessibility and the seniority of exploration work can combine to produce a denser concentration zone than a purely random distribution of the resource would suggest. This reading nevertheless remains an initial spatial approach: the map should be interpreted as an exploratory prospecting tool, intended to direct attention toward certain sectors, and not as a demonstration of an exploitable deposit.

8. Limitations

Several limitations should be noted before concluding. The exploratory score relies on criteria chosen a priori, whose weighting remains relatively simple and does not necessarily reflect the relative importance that specialized geologists would assign to each of them. Keyword-based filtering, although improved after correcting the “Li” false positive, remains sensitive to the exact wording of the descriptions contained in the database, which can lead to omitting some relevant targets or retaining others only marginally.

The distance calculation used for cross-referencing occurrences and targets relies on an approximate geographic distance between coordinates. It does not account for terrain relief, actual ground accessibility, roads, or logistical constraints. At the cartographic scale used in this report, the entirety of Quebec's territory is represented in a relatively compact way, which can give an impression of proximity between points that are in reality separated by several tens of kilometers of poorly accessible territory.

Finally, and this is the most fundamental limitation, this work does not rely on any drilling data, any detailed geochemical analysis, nor any study of bedrock geology. Only field-based geological expertise, drawing on data far more detailed than that available in SIGÉOM in tabular form, would make it possible to truly conclude on the existence of exploitable mining potential in the areas of interest identified here.

9. Conclusion

This report has proposed an exploratory reading of lithium in Quebec based on public SIGÉOM data. The analysis identified nine lithium projects, seven of which are located within the Plan Nord perimeter, then broadened the scope to 159 GMLI occurrences distributed across three potential classes built from an exploratory score. Cross-referencing these occurrences with 44 exploration targets retained after correcting the keyword filtering showed that more than half of them, 91 out of 159, are located less than 100 km from a target already flagged by geologists.

The final mapping highlighted a non-uniform spatial organization of these signals, concentrated around a few already known mining poles, particularly in the northwestern sector associated with the Galaxy, Rose Lithium-Tantale, and Whabouchi projects. At Quebec's scale, this concentration illustrates the classic mechanisms of economic geography: the location of mining activities depends not only on the presence of the resource, but also on territorial accessibility, the history of exploration, and the strategies of promoters already engaged in the region.

The value of this approach lies above all in its method: it shows that it is possible, using open data and relatively accessible tools, to build a coherent initial spatial reading of a contemporary mining issue, without specialized geological expertise. The results obtained nevertheless remain favorable signals rather than proof, and naturally call for further work: integrating drilling and detailed geochemical data, accounting for bedrock geology, or analyzing the actual accessibility of identified sectors based on their distance to existing roads and infrastructure networks.

Data Sources

All the data used in this report comes from SIGÉOM, Quebec's public geo-mining database managed by the Ministère des Ressources naturelles et des Forêts, accessed via the Géologie Québec portal. Three databases from this platform were used: “Mines and Projects,” “Metallic Substances,” and “Exploration Targets.”

- SIGÉOM / Géologie Québec — Quebec's public geo-mining data, “Mines and Projects,” “Metallic Substances,” and “Exploration Targets” databases: <https://gq.mines.gouv.qc.ca/documents/SIGEOM/TOUTQC/FRA//>
- Géologie Québec: <https://gq.mines.gouv.qc.ca/>
- Natural Earth — “Admin 1 — States, Provinces” administrative layer, used for Quebec's boundary outline: <https://www.naturalearthdata.com/>
- OpenStreetMap — basemap used in QGIS: <https://www.openstreetmap.org/>

Extraction, filtering, construction of the exploratory score, and distance calculations were carried out independently using Python. The mapping was produced in QGIS, based on the layers mentioned above.

Appendix — Cross-Referenced Ranking of the Ten Highest-Scoring Occurrences

The following table presents, for the ten occurrences with the maximum exploratory score, the nearest exploration target, the corresponding distance, and the associated proximity class. Some occurrence names appear more than once, as they may correspond to distinct mineralized bodies or distinct records in the SIGÉOM database.

Occurrence	Score	Nearest Target	Distance (km)	Proximity Class
Adina Main (Jamar)	14	(6) Adina (14-AB-6048A)	1.01	Very close to a target
CV13	14	10-SB-6326	11.85	Very close to a target
CV5	14	10-SB-6326	15.09	Very close to a target
CV5	14	10-SB-6326	15.60	Very close to a target
CV5	14	10-SB-6326	16.55	Very close to a target
Cyr-Lithium (James Bay Lithium)	14	Béryl	38.11	Close to a target
Rose	14	FS-3292	68.27	Within a 100 km radius
Whabouchi	14	FS-3292	103.33	Far from targets
Authier	14	Indice Kekek (15-AM-200B)	170.54	Far from targets
Lac Moblan Ouest	14	PR-6073	219.24	Far from targets